

BIKRANT NATH

+91 9101121475 | bikrantnath4@gmail.com | Guwahati, Assam (open to relocation)
linkedin.com/in/bikrant-nath-7747781a7 | github.com/bikrantnath22 | bikrant-nath.netlify.app

SUMMARY

AI/ML engineer finishing my M.Tech in CSE in 2026. I like building LLM apps that work end to end, and I have already taken a full-stack product live with real customers.

EDUCATION

M.Tech, Computer Science & Engineering | Tezpur University | CGPA 7.63/10

Aug 2024-Jul 2026

EXPERIENCE

M.Tech Thesis Researcher | Tezpur University | GitHub

Aug 2025-Jul 2026

DP-CTGAN: Differentially Private Synthetic Data Generation with Rare Minority Class Sampling | Guide: Dr. S Ibotombi Singh, Asst. Prof.

- Developed DP-CTGAN to fix a known problem with private synthetic data: privacy noise wipes out rare cases like fraud. Rare-class sampling keeps them intact at $\epsilon = 3$.
- Evaluated on four datasets (Credit Card Fraud, Adult Census, Taiwan and MIMIC-III), and a paper is currently in preparation.
- On highly imbalanced Credit Card Fraud dataset(0.17% fraud cases) it achieved an AUPRC of 0.7933 and an F1 of 0.7512 with almost no membership-inference leakage, and along the way I caught and fixed an AUC-inflation bug in my own evaluation

Tech Stack: Python, PyTorch, CTGAN, Opacus DP-SGD, Rényi accountant, normalizing flows, Optuna, Pandas, Scikit-learn

Co-Founder & Full-Stack Developer | Athaana, grocery eCommerce, Nalbari | GitHub

Feb 2022-Jun 2022

- I built the whole platform myself: listings, cart, checkout, JWT login and live order tracking, hosted on DigitalOcean.
- We launched with 4 people , got customers via Facebook Ads and SEO , and shipped over 250+ real orders before closing.

Tech Stack: React , Node.js, Express.js, MongoDB , Tailwind CSS, JWT, DigitalOcean

Software Development Intern | Fineworks Media & Solution Technologies, Guwahati

Jan 2022-Jul 2022

- Wrote the REST APIs behind a hotel booking app, including role-based JWT login and search by date, price and availability.
- Setup schemas for MongoDB and added checks so a room can't be double-booked,also added PayPal payments and webhooks.

Tech Stack: Node.js ,Express.js, MongoDB ,React ,REST APIs ,JWT ,PayPal API

PROJECTS

AI Research & Fact-Verification Agent | Live Demo | GitHub

2026

- LLM answers often sound confident, but cite nothing Built a 4-agent LangGraph pipeline (Planner, Retriever, Synthesiser, Verifier) with inline citations.
- Qdrant vector search + live Tavily search. 3-signal hallucination risk score (semantic entropy, ensemble disagreement, faithfulness) triggers re-querying.
- Exposed as 3 MCP servers; deployed as 5 service Docker Compose stack on AWS EC2 + Nginx via GitHub Actions CI/CD. Includes unit and integration tests.

Tech Stack: Python, LangGraph, FastAPI, Groq, Qdrant, Tavily, MCP, DeBERTa, Docker, GitHub Actions, AWS EC2

RAG Guardrail System | GitHub

2026

- To stop prompt injection reaching a RAG app, I built a four-layer input guard (regex, DeBERTa, MiniLM, LLM judge).
- A two-layer output guard redacts PII based on role, and three access tiers (Guest, Employee, Admin) enforce role-based access control (RBAC).
- Three real attack scenarios with a Dockerized dashboard visualising the guards to prevent data exfiltration.

Tech Stack: Python ,FastAPI ,PyTorch , DeBERTa ,Sentence-Transformers,Hugging Face ,Groq ,Docker

Agentic Hybrid RAG & NLI Citation Verifier | GitHub

2026

- Developed a hybrid RAG pipeline combining Dense & BM25 search (RRF) with cross-encoder reranking, and an autonomous query-rewrite loop.
- Hallucinations were removed with a stringent DeBERTa NLI verifier, while performance was validated by automated Recall@K and MRR metrics.

Tech Stack: Python, Groq API, HuggingFace Transformers ,DeBERTa NLI, NumPy , BM25,Pandas

TECHNICAL SKILLS

Languages & Libraries: Python , JavaScript,SQL , PyTorch, Scikit-learn , Hugging Face Transformers ,NumPy ,Pandas,

ML & Deep Learning: Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), Tokenization, Embeddings,Transformers, GANs, Differential Privacy

GenAI & Agents: Large Language Models (LLMs), LangChain, LangGraph, RAG, Llamaindex,Vector Databases,Prompt Engineering, Model Context Protocol (MCP), Agentic Workflows

MLOps & Backend: FastAPI, Node.js, REST APIs, GitHub Actions (CI/CD),Docker, Kubernetes, AWS, Nginx, Linux, Git, React, MongoDB, MySQL

Core CS: Data Structures & Algorithms (DSA) , DBMS , Object-Oriented Programming (OOP) ,Operating Systems

CERTIFICATIONS & ACHIEVEMENTS

- NPTEL certifications:** Machine Learning , Cloud Computing , IoT , Ethical Hacking Certificates Link
- 1st Runner-Up, Codestellation Hackathon 2022**(Assam Engineering College): live-preview resume builder, top 2 of 25+.
- NGIN seed funding**, INR 1,00,000 for a VR-based computer assisted intervention project (2024).